

GIZMO Operations Guide

Never lift the safety ground. Follow SOP004 and the current work plan.

Installed system

GIZMO is already deployed. SOP004 authorizes its installation and use. The initial grounding setup will use one saturable inductor.

Item	Recorded setting
Excitation	1436 Hz
Runtime / OPC UA model	0.5.1 / 1.4.0
Monitoring server	dune-fd-test01
History available	Since July 27, 2026

These are stored settings. The server dashboard and historian are running; remote measurements are unavailable.

Connections after installation or maintenance

With the rack de-energized, check each cable against its installed label and the approved drawing:

- Pearson CT output to GIZMO CT input; the CT observes the intended safety-ground path.
- Stimulus center to the designated Detector Ground point; return to the designated rack/infrastructure reference.
- Beacon and audible output to their labeled connectors.
- Network to the assigned controls port; approved DC supply to the chassis.
- One saturable inductor in the initial setup, with its protection and safety-ground connections verified by the electrical lead.

Changes to grounding, CT routing, stimulus, or calibration require the applicable work controls.

Contractor shift checks

Contractor personnel work with the cryostat and beam structure connected to Building Ground. At shift start and handover, acknowledge the grounding status:

1. Check the Building Ground bond with a multimeter.
2. Check the knife-switch position and tagout status.
3. Clear tools and electrical cords so the FNAL I&I/GIZMO group can test before, after, or between shifts.

GIZMO readings, dashboard checks, and isolation tests are the responsibility of the FNAL I&I/GIZMO group.

FNAL I&I/GIZMO isolation tests

Use the planned knife-switch/tagout arrangement for periodic tests of shorts, capacitance, and coupling between Detector Ground and Building Ground.

1. Coordinate the test window with the contractor and confirm the area is clear of tools and electrical cords.
2. Establish and record the test grounding state using the authorized switching/tagout procedure.
3. Check fresh data, measurement quality, and frequency. Record isolation, valid capacitance, and coupling diagnostics; compare with the permitted limits.
4. Restore and verify the required Building Ground connection, complete tagout closeout, and hand the structure back before contractor work resumes.

Plan tests before, after, or between shifts, often enough to avoid disruptive late discovery of unwanted connections. Test cadence, detailed shift routine, settling time, permitted isolation limits, and failed/missed-test response: TBD.

Record test time, grounding state, findings, and handback in Procore Daily Log > GIZMO. HIGH Z is a range indication, not a resistance value. The last stored sample had Uncertain quality; it does not establish the present ground condition.

Measurement availability requirement

As of September 30, 2026, remote GIZMO measurements have been unavailable for 17 days, following loss or damage of the mine network switch. The last retained sample is September 13 at 07:11:34 UTC.

The facility must provide technical support to keep GIZMO measurements available throughout cryostat installation. Support must cover the instrument, mine network switch, communications, and measurement recovery.

When remote measurements are unavailable, the FNAL I&I/GIZMO group must provide local measurements in their place and record the results. Contractor work must not be delayed by facility monitoring outages; the facility must maintain the local measurement alternative.

Assign the support contact, local measurement method, test cadence, and recovery time: TBD.

FNAL: server check

On dune-fd-test01, run:

```
/storage/gizmo-monitor/bin/status
```

It reports the historian, dashboard, upstream connection, stored samples, and disk space. Check upstream connectivity and the newest GIZMO sample, not just whether the processes are running.

For a local dashboard connection:

```
ssh -N -L 127.0.0.1:18081:127.0.0.1:18081 dune-fd-test01
```

Then open <http://127.0.0.1:18081/> in the browser. Use the configured SSH access for the server.

FNAL: shutdown

Coordinate any loss of monitoring before shutdown. Use SOP004 and the deployed operating procedure to disable excitation and power down. Leave the permanent safety-ground path intact. Record the time and reason.